BT604 FINALTERM PAPER

Mcqs

1. **White Biotechnology also known as industrial biotechnology** applies to industrial processes. Designing of an organism to use to produce a useful chemical.

2. Products which are produced by recombinant DNA technology (produced by biotechnology)
   Antibiotics
   Vaccines
   GMO's
   Transgenic plants
   Baverages

3. **Bioinformatic**s is the use of computing technologies to manage and analyse biological data.

4. E. Coli was discovered in **1885** by the German bacteriologist Theodor escherich.

5. The cytoplasmic matrix is maintained at ph **7.6–7.8**

6. **Ascomycetes** is the largest class of fungi and includes the yeasts that are utilized in many industrial fermentation processes

7. **Pilli** definition

8. **Mitochondria** Fully developed organelles are spherical to rod-shaped, enclosed by a double membrane.

9. Microorganisms may contain more than **15% (w/w) nitrogen**, mostly within structural and functional proteins, and nucleic acids.

10. During batch fermentations the population of microorganisms goes through several distinct growth phases: **lag, acceleration, exponential growth, deceleration, stationary and death**

11. In spread plating, samples, usually **0.1ml**, are spread on the surface of a suitable agar-based nutrient medium using a sterile spreading device

12. **obligate aerobes** grow only in the presence of oxygen, as it is required as the terminal electron acceptor for electron transport in aerobic respiration

13. The break down of complex molecules into simpler ones with the release of energy is called **catabolism**

14. glucose has a repressive effect on the production of many **secondary metabolites,** including antibiotics(b-lactams and tetracylines) and some alkaloids.

15. **Conjugation** involves cell-to-cell contact, where the donor contacts the recipient with a filamentous protein structure called a sex pilus,

16. The first step requires total DNA extraction from the donor organism, which is then cut into smaller sequences using a specific **restriction endonuclease.**

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17. Introduction of recombinant plasmids into bacterial cells can be achieved following **calcium chloride** treatment, which renders the cell membranes more permeable to DNA

18. Maize starch is most widely used, but it may also be obtained from other cereal and root crops.

19. Whey is an aqueous by product of the dairy industry.

20. Methanol has a high percent carbon content and is relatively cheap, although only a limited number of organisms will metabolize it.

21. PNEUMATIC SYSTEMS, Such as airlift fermenters, have no Moving parts and use the expansion of compressed gas to bring about the mixing

22. The rate of use by the biological system will be determined by the affinity and saturation characteristics of the **terminal oxidase**.

23. Some industrial fermentations are **not aseptic**, but microbial contamination is still maintained at a minimum level by boiling or pasteurization of the media.

24. Broth conditioning techniques are mostly used in association with **sedimentation and centrifugation** for the separation of cells from liquid media.

25. the Manton and Gaulin homogenizer (APV-type mill), are employed for **pilot- and production-scale** cell disruption.

26. standard operating procedures (SOPs)

27. Several microbial enzymes are employed in fruit juice processing, but probably the most important are **pectinases.**

28. Ethanol is an attractive fuel because it may be used alone or mixed with other liquid fuels, e.g. 'gasohol', a blend of **10–22% (v/v) ethanol** with gasoline

29. alcoholic beverages, especially beers, wines and distilled spirits, which are derived from **sugars and starches**

30. The term beer is given to **non-distilled alcoholic** beverages made from partially germinated cereal grains, referred to as malt.

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- Factors for oxygen transfer to the medium of fermenter 2
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- Why foam is produced and what problem is caused during fermentation 3
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